

Belief Revision in Truthmaker Semantics

Shin JeongBean

September 27, 2026

Abstract

In this paper, I propose several hyperintensional frameworks for belief revision by using resources from Finean truthmaker semantics, resembling the traditional AGM approach. Despite its exquisiteness and utility, one may desire alternative frameworks for belief revision, which decide a definite contraction given the initial epistemic state, free from logical omniscience, classical consistency assumptions and intensionality, while avoiding the problematic cases of the recovery postulate. Finally, one might have a broad desideratum that the modelled epistemic agent must be rational only within a local context.

I develop some frameworks satisfying those desiderata by shifting the locus of doxastic operations from classical propositional belief sets to belief-state sets, which comprise exact or inexact verifiers (truthmakers) of the formulas toward which the modelled agent has positive doxastic attitude. I first present two simple frameworks which satisfy all of the desiderata except local rationality, and move forward to develop some frameworks which satisfy local rationality through varying degrees and formal mechanisms. Specifically, I introduce a subject-matter locality constraint to accommodate the problem of unrestricted introduction of disjunctions, and secure minimal rationality either by imposing consistency constraints on belief-state sets, or a weighing mechanism that balances verifiers against falsifiers.

1 Introduction

In this paper, I will develop a family of hyperintensional frameworks for belief revision utilizing the resources provided by Kit Fine's truthmaker semantics. The classical AGM framework (notably Alchourrón, Gärdenfors, and Makinson, 1985; the description below follows that of van Ditmarsch, van der Hoek, and Kooi, 2008) represents an agent's doxastic state as a theory or belief set $\mathcal{K}$ in a propositional language. Let $\mathcal{L}$ be the propositional language generated by a countable set of atomic propositions P under the classical truth-functional connectives. Let Cn be the classical consequence operator, where $\text{Cn}(\Sigma) = \{\phi \in \mathcal{L} \mid \Sigma \vdash \phi\}$. A *belief set* $\mathcal{K} \subseteq \mathcal{L}$ is defined as a set of formulas closed under Cn , i.e., $\mathcal{K} = \text{Cn}(\mathcal{K})$. The unique inconsistent belief set is denoted by $\mathcal{K}_\perp = \mathcal{L}$.

The AGM framework includes three basic operations that map a belief set $\mathcal{K}$ and an input sentence $\phi \in \mathcal{L}$ to a new belief set. Intuitively,

1. **Expansion** ($\mathcal{K} \oplus \phi$): The incorporation of new information ϕ into $\mathcal{K}$, without retracting any existing beliefs.
2. **Contraction** ($\mathcal{K} \ominus \phi$): The removal of a belief ϕ from $\mathcal{K}$, such that the resulting set no

longer implies ϕ , unless ϕ is a classical tautology ($\vdash \phi$).

3. **Revision ($\mathcal{K} \circledast \phi$):** The integration of a new, potentially conflicting belief ϕ into $\mathcal{K}$ (if ϕ is consistent), while some sentences in $\mathcal{K}$ may need to be removed so the resulting set is also consistent.

The specific behavior of expansion is characterized by the postulates $(\oplus 1)$ - $(\oplus 6)$ in Table 1. This additionally captures the idea that $\mathcal{K} \oplus \phi = \text{Cn}(K \cup \{\phi\})$.

The postulates for expansion determine a clear and unique result after expansion, whereas characterizing contraction is more complicated under the AGM approach. Here I will not go deeper beyond so-called *partial meet contraction*, which conforms to the postulates $(\ominus 1)$ - $(\ominus 6)$ in 2. The idea behind it is roughly that if ϕ is contracted from $\mathcal{K}$, the resulting belief set is the intersection of the favored maximal subsets of $\mathcal{K}$ that do not imply ϕ . If ϕ is not in $\mathcal{K}$, then the maximal subset of $\mathcal{K}$ that does not imply ϕ would be $\mathcal{K}$ itself.

$(\oplus 1)$ $\mathcal{K} \oplus \phi$ is a belief set	<i>type</i>
$(\oplus 2)$ $\phi \in \mathcal{K} \oplus \phi$	<i>success</i>
$(\oplus 3)$ $\mathcal{K} \subseteq \mathcal{K} \oplus \phi$	<i>expansion</i>
$(\oplus 4)$ $\phi \in \mathcal{K} \implies \mathcal{K} = \mathcal{K} \oplus \phi$	<i>minimal action</i>
$(\oplus 5)$ For all $\mathcal{H}, \mathcal{K} \subseteq \mathcal{H} \implies \mathcal{K} \oplus \phi \subseteq \mathcal{H} \oplus \phi$	<i>monotony</i>
$(\oplus 6)$ $\mathcal{K} \oplus \phi$ is the smallest set satisfying $(\oplus 1) - (\oplus 5)$	<i>minimal change</i>

Table 1: The AGM Postulates for Expansion.

$(\ominus 1)$ $\mathcal{K} \ominus \phi$ is a belief set	<i>type</i>
$(\ominus 2)$ $\mathcal{K} \ominus \phi \subseteq \mathcal{K}$	<i>contraction</i>
$(\ominus 3)$ $\phi \notin \mathcal{K} \implies \mathcal{K} = \mathcal{K} \ominus \phi$	<i>minimal action</i>
$(\ominus 4)$ $\not\vdash \phi \implies \phi \notin \mathcal{K} \ominus \phi$	<i>success</i>
$(\ominus 5)$ $\phi \in \mathcal{K} \implies \mathcal{K} \subseteq (\mathcal{K} \ominus \phi) \oplus \phi$	<i>recovery</i>
$(\ominus 6)$ $\vdash \phi \leftrightarrow \psi \implies \mathcal{K} \ominus \phi = \mathcal{K} \ominus \psi$	<i>extensionality</i>

Table 2: The Basic AGM Postulates for Contraction.

Once expansion and contraction are characterized, we can define revision in terms of these two operations by so-called *Levi Identity*:

$$\text{Levi Identity: } \mathcal{K} \circledast \phi = (\mathcal{K} \ominus \neg \phi) \oplus \phi$$

The revision operation, defined with the expansion and the partial meet contraction conforming to the postulates above can be axiomatized via the postulates $(\circledast 1)$ - $(\circledast 6)$ in Table 3.

$(\circledast 1)$ $\mathcal{K} \circledast \phi$ is a belief set	<i>type</i>
$(\circledast 2)$ $\phi \in \mathcal{K} \circledast \phi$	<i>success</i>
$(\circledast 3)$ $\mathcal{K} \circledast \phi \subseteq \mathcal{K} \oplus \phi$	<i>upper bound</i>
$(\circledast 4)$ $\neg \phi \notin \mathcal{K} \implies \mathcal{K} \oplus \phi \subseteq \mathcal{K} \circledast \phi$	<i>lower bound</i>
$(\circledast 5)$ $\mathcal{K} \circledast \phi = \mathcal{K}_\perp \iff \vdash \neg \phi$	<i>triviality</i>
$(\circledast 6)$ $\vdash \phi \leftrightarrow \psi \implies \mathcal{K} \circledast \phi = \mathcal{K} \circledast \psi$	<i>extensionality</i>

Table 3: The AGM Postulates for Revision.

It would be redundant to spill more ink to praise the exquisiteness and the usefulness of the AGM approach, since it has already been appreciated for several decades. Yet for some reason one might desire different kinds of frameworks. In section 1.1, I will propose the desiderata of the framework I will develop in this paper. Section 1.2 will be devoted to deploying the conceptual materials from Finean truthmaker semantics. Section 2 presents two simple frameworks that partially satisfy the desiderata in section 1.1. Several slightly more sophisticated frameworks, which are stronger than those presented in §2, are developed in section 3, which fully satisfy the desiderata with different degrees and means.

1.1 Desiderata

1.1.1 Definite Contraction

The AGM framework assumes unique contraction. That is, by $(\ominus 1)$, the result of contracting ϕ from $\mathcal{K}$ must be one particular belief set. But the postulates $(\ominus 1)$ - $(\ominus 6)$ by themselves do not guarantee that there is a unique belief set that satisfies them.

For example, suppose $\mathcal{K} = \text{Cn}(\{p, q, r\})$. In this case $\mathcal{K} \ominus (p \wedge q)$ has multiple possible outcomes. By $(\ominus 4)$ and the definition of belief set, it would be evident that at least one of p and q should be removed from the belief set; but which one? Either choice satisfies the postulates, and even if one removes both of them, it is still a valid partial meet contraction. But then, how do we choose one among them? In partial meet contraction, the choice was somewhat arbitrary.

The standard way out of the arbitrariness was to incorporate two more postulates $(\ominus 7)$ and $(\ominus 8)$ - which results in *transitively relational partial meet contraction*. This turned out to capture the idea of *epistemic entrenchment* (Gärdenfors & Makinson, 1988). In cases where the epistemic agent is more entrenched into p , $\mathcal{K} \ominus (p \wedge q)$ would be $\text{Cn}(p, r)$, whereas in cases where it is more entrenched into q , the result would be $\text{Cn}(q, r)$.

$(\ominus 7) ((\mathcal{K} \ominus \phi) \cap (\mathcal{K} \ominus \psi)) \subseteq \mathcal{K} \ominus (\phi \wedge \psi)$	<i>min-conjunction</i>
$(\ominus 8) \phi \notin \mathcal{K} \ominus (\phi \wedge \psi) \implies \mathcal{K} \ominus (\phi \wedge \psi) \subseteq \mathcal{K} \ominus \phi$	<i>max-conjunction</i>

Table 4: Additional AGM Postulates for Contraction.

Yet one could still complain that the entrenchment cannot be inferred from the belief set alone. The transitively relational partial meet contraction forces a regularity on contracting some formulas from $\mathcal{K}$, but still does not by itself decide what the regularity is. The regularity can be determined only if the entrenchment is already specified. But to fully specify the denumerably long list of entrenchment is, for one thing, practically not very realistic. Thus, one might desire a framework which can determine the result of contraction simply from what is contracted from what, just as expansion in the AGM framework is definite in that sense.

A related consideration is that one might want to model the epistemic entrenchment as a subject of change as the belief set undergoes some operations. Let $\mathcal{K} = \text{Cn}(\{p, \neg q\})$. It seems natural to me to suppose that if $\mathcal{K}$ is revised with $p \wedge q$, not only is $\neg q$ contracted, but also that p is possibly in some sense *reinforced*, so p might be more entrenched to the agent after the operation of revision. Thus, one might desire entrenchment to be understood in terms of operations such as expansion and contraction; not to determine the result of contraction via entrenchment.

So this gives the first desideratum for frameworks I will develop in this paper:

- (D1) **Definite Contraction:** *The definition, or the postulates of the contraction operator alone should be sufficient to determine the result after the contraction*

1.1.2 Logical Ignorance, Paraconsistency, and Local Rationality

Logical omniscience is a traditional worry for any kind of epistemic logic with the necessitation rule: $\vdash \phi \implies \vdash \Box \phi$. In epistemic logic, this gives that an epistemic agent has the doxastic or epistemic attitude which is supposed to conform to the behavior of $\Box$, towards all logical truths. Usually this yields the consequence that an ideal epistemic agent must know or believe all logical truths, which may seem suspicious.

The AGM approach yields the same result. Since the belief set $\mathcal{K}$ is defined as a set of formulas closed under the classical consequence operator, in any case the epistemic agent believes all the logical truths, which causes the same discomfort as the problem of logical omniscience for epistemic logics.

An additional worry is that logical omniscience results in dismissing the contraction with logical truths. This is what the *minimal action* postulate commands; but even without the postulate the definition of $\mathcal{K}$ itself forbids the removal of any logical truths. Revising *minimal action* with a weaker postulate at most only results in eccentric possibilities where, by contracting a logical truth, it rather takes out some innocent formulas other than what is contracted. However, one might want to model what an ideally rational epistemic agent must believe after it is forced to contract a logical truth from the belief set.

Likewise, one might feel uncomfortable with the idea that one must believe all the logical consequences of its belief in general. One case we might want to model is the pessimism about one's own belief. One might be equally deeply entrenched in each of the 'atomic' beliefs, but come to contract the conjunction of all of its beliefs, since it also has some certainty that at least some of its beliefs would be wrong?that the agent itself cannot be perfect. In this case, a faithful modelling seems not to remove all the beliefs (or all the logically intermediate beliefs) from the original belief set, but only to remove the conjunction, yet retaining all the 'atomic' beliefs. But to allow this, one must admit a belief set that is not closed under classical consequence.

So this gives the second desideratum:

- (D2) **Logical Ignorance:** *The belief set must be able to exclude its logical consequences*

There is also a similar worry for expansion. Again, since $\mathcal{K}$ is closed under classical consequence, starting with inconsistent beliefs, or expanding with any ϕ such that $\mathcal{K} \vdash \neg \phi$ results in explosion.

But it may be unclear why an ideally rational epistemic agent should be so vulnerable to inconsistency. Especially, one might want to model how one should operate when the initial state of the belief set is reasonably inconsistent. This gives the third desideratum:

- (D3) **Paraconsistency:** *The belief set must be able to admit inconsistency without explosion*

Finally, the motivating idea behind **Logical Ignorance** and **Paraconsistency** may be the consideration that we might want to model an epistemic agent that is *locally* rational, in the sense that it contemplates a subset of the whole belief set at a time. Indeed, human epistemic agents never inspect every corner of our own belief set when expanding or contracting our beliefs;

and this is somewhat desirable even for an ideal epistemic agent when one considers cognitive economy.

This is also motivated by the idea that the strength of those two desiderata must be somehow inhibited. For example, **Paraconsistency**, in its extreme, would allow one to believe apparent contradictions such as $\phi \wedge \neg\phi$; yet we might want the modelled ideal epistemic agent to be aware enough not to make such a silly mistake. **Logical Ignorance**, on the other hand, would make it possible that the ideal agent believes both ϕ and $\neg\phi \vee \psi$ ($\iff \phi \rightarrow \psi$) without believing ψ . Although we might be reluctant to take the modelled agent to be perfect in all logical regards, we still want it to be locally *rational*.

So this gives the fourth, somewhat vague desideratum:

(D4) **Local Rationality**: *The modelled epistemic agent must be rational, but only locally so.*

1.1.3 Hyperintensionality

In §1.1.2, it was suggested that the new framework must be able to contract with logical truths. Naturally, if it is possible to contract with logical truths, it would also seem possible to expand or revise with logical truths.

Yet with *extensionality* postulates for contraction and revision, non-trivial contraction and revision with logical truths consist only of two kinds: either you contract *all* logical truths, or you revise and thus incorporate *all* logical truths. Suppose a belief set $\mathcal{K}$ such that $\{p \vee \neg p, q \vee \neg q\} \subset \mathcal{K}$ and $r \vee \neg r \notin \mathcal{K}$. This would be possible in frameworks where **Logical Ignorance** is satisfied. Now consider $\mathcal{K} \ominus (p \vee \neg p)$. Since $\vdash (p \vee \neg p) \leftrightarrow (q \vee \neg q)$, by *extensionality* of contraction $\mathcal{K} \ominus (p \vee \neg p) = \mathcal{K} \ominus (q \vee \neg q)$, and by *success* we get $\{p \vee \neg p, q \vee \neg q\} \cap \mathcal{K} \ominus (p \vee \neg p) = \emptyset$. Furthermore, by extensionality for revision, likewise we get $\{p \vee \neg p, q \vee \neg q, r \vee \neg r\} \subset (\mathcal{K} \ominus (p \vee \neg p)) \circledast (r \vee \neg r)$.

But one might desire a more fine-grained contraction and revision. For one thing, if one already has **Local Rationality**, one might want to be able to contract some particular kind of logically true formulas without affecting other formulas, especially for those that have different *subject matter*. Likewise, one might want to be able to expand, or revise with some logically true formulas without bringing infinities of logical truths into the belief set.

So this gives the fifth desideratum:

(D5) **Hyperintensionality**: Two extensionally, or intensionally equivalent but hyperintensionally distinct formulas must be able to behave differently upon expansion or contraction.

1.1.4 Pertinent Recovery

Finally, it should be noted that the *recovery* postulate was a subject of dispute for a long time. The idea that if one loses a belief and immediately regains it, the agent must at least return to the original belief set would seem intuitively compelling. The postulate also plays a crucial role in preventing the contraction from being unnecessarily strong, so contracting with ϕ cannot remove more formulas from the belief set than what expanding with ϕ would take into the belief set.

Yet *recovery*, as the AGM framework postulates, also yields counterintuitive results. Suppose $\{p, q, p \wedge q, p \vee q\} \subset \mathcal{K}$. Now, the fact that $\{p, q, p \wedge q, p \vee q\} \cap (\mathcal{K} \ominus (p \vee q)) = \emptyset$ will follow

in the AGM framework (by *success* of contraction and the definition $\mathcal{K} = \text{Cn}(\mathcal{K})$), and it is quite intuitive. If the modelled epistemic agent loses the belief that it is either the case that p or q , reasonably the agent would also lose the belief that it is both the case that p and q . So far so good, but *recovery* yields that if the modelled agent immediately regains the belief that $p \vee q$, the agent must also regain the belief that $p \wedge q$. That is: $\{p, q, p \wedge q, p \vee q\} \subset (\mathcal{K} \ominus (p \vee q)) \oplus (p \vee q)$. But this is intuitively not very compelling.

To generalize: suppose $\{\phi, \psi\} \subset \mathcal{K}$ where ϕ strictly entails ψ but not the other way around. This yields the result that $\phi \notin \mathcal{K} \ominus \psi$. Yet by recovering ψ , the modelled agent recovers not only ψ but the stronger ϕ . This is also suspicious, for recovering ψ seems to have no reason to involve ϕ if ψ does not entail ϕ , regardless of the fact that ϕ was in the initial belief set before the contraction (Hansson, 2022).

It seems to me that the intuition behind proposed counterexamples for *recovery* is that one might have different *sources* for the same belief. If ψ is to be contracted from $\mathcal{K}$, not only should the *sources*, or reasons for believing ψ be discarded, but also should the sources of the belief that ϕ be defeated, for any source of the belief that ϕ would also be the source of the belief that ψ . But when one recovers the belief that ψ , it is not guaranteed that the recovery is via reconfirming the exact source for both beliefs that ψ and ϕ . One may recover the belief that ψ via a *new* source for the belief that ψ which does not make one believe that ϕ . It is possible because, since ψ does not entail ϕ , the source for the belief that ψ is not necessarily the source for the belief that ϕ .

Notice how *recovery* is extremely compelling when it is formulated in terms of the source of belief: if a source of belief is once discarded and immediately reconfirmed, the original belief set must be a subset of the belief set after the sequence. Thus, one might wish the framework to be able to grasp this sense of recovery, not the one among the AGM postulates.

This gives the final, sixth desideratum:

(D6) **Pertinent Recovery:** The framework must avoid the problematic case of *recovery*.

1.2 Materials From Truthmaker Semantics

In this paper, I propose frameworks that satisfy **Definite Contraction**, **Logical Ignorance**, **Paraconsistency**, **Hyperintensionality**, and **Pertinent Recovery**. **Local Rationality** will not be satisfied in two prototypes, and will be realized in other frameworks in different degrees.

The core idea is to posit a *belief-state set* $\mathcal{S}^{\mathcal{K}}$, where any belief-state set determines a unique belief set. Intuitively, $\mathcal{S}^{\mathcal{K}}$ is the set of the *sources* of the modelled agent's belief; so any member of $\mathcal{S}^{\mathcal{K}}$ is a state which *verifies*, or *truthmakes* some member of $\mathcal{K}$. Operations are defined to operate on $\mathcal{S}^{\mathcal{K}}$ instead of $\mathcal{K}$; thus we expand, contract or revise with a state σ from a belief-state set $\mathcal{S}^{\mathcal{K}}$. The expansion, contraction or revision with a formula ϕ from a belief set $\mathcal{K}$ is only defined in terms of the operation with a state on a belief-state set.

The ideas presented in this subsection originate from Kit Fine's work on truthmaker semantics. Below in this subsection I introduce some important resources from Finean truthmaker semantics which will be directly relevant to new frameworks developed in this paper. The following details are mostly taken from Fine (2016), Fine (2017a), Fine (2017b), and Fine (2017c).

A state, informally, is anything that exactly verifies or exactly falsifies a statement (the no-

tion of exact and inexact verification/falsification will be covered shortly). We do not need to assume a particular ontology of states to do semantics, as we do not need to assume a particular ontology of possible worlds to do modal logic.

One characteristic feature of states is that states can have mereological structure. Thus a state can be a part of another state; and any two states can fuse to compose a single state. The parthood relation here is supposed to be antisymmetric and transitive.

Formally, we achieve such an understanding of states by defining a *set space*. To do so we first define the parthood relation $\sqsubseteq$ and the notion of *least upper bound*.

Definition 1.1 (Parthood Relation on S). $\sqsubseteq$ (*part*) is a binary relation on a set of states S which conforms to the following three conditions:

$$\begin{aligned} \text{Reflexivity: } & \forall s(s \sqsubseteq s) \\ \text{Antisymmetry: } & \forall s \forall t((s \sqsubseteq t \& t \sqsubseteq s) \implies s = t) \\ \text{Transitivity: } & \forall s \forall t \forall u((s \sqsubseteq t \& t \sqsubseteq u) \implies s \sqsubseteq u) \end{aligned}$$

Definition 1.2 (Least Upper Bound). A state s is an upper bound of $T \subseteq S$ iff $\forall t \in T \ t \sqsubseteq s$. A state s is a least upper bound of T iff s is an upper bound of T , and it is part of any upper bound s' of T .

Definition 1.3 (State Space). A state space is an ordered set $\langle S, \sqsubseteq \rangle$, where S is a set of states and $\sqsubseteq$ (*part*) is a binary relation on S , and for any $T \subseteq S$ its least upper bound exists.

It will be easy to check that the least upper bound is unique by antisymmetry. The least upper bound (sometimes called ‘fusion’) of T is denoted by $\sqcup T$, or $t_1 \sqcup t_2 \sqcup t_3 \dots$ for $T = \{t_1, t_2, t_3, \dots\}$. Also, it would be easy to check that if $s' \sqsubseteq s$, $s' \sqcup s = s$, since s and s' are both part of s and all upper bounds of $\{s', s\}$ must by definition have s as part.

Two extreme states are worth noting:

Convention 1.1 (Null State and Full State). Let ‘ $\square$ ’ denote the least upper bound of $\emptyset \subseteq S$, which is a part of any state and has no part. This will also be called ‘null state’. Let ‘ $\blacksquare$ ’ denote the least upper bound of $S \subseteq S$, which is not a part of any state and has every state as part. This will also be called ‘full state’.

The propositional language we will use throughout the paper is defined as follows:

Definition 1.4 (Propositional Language $\mathcal{L}$). Let P be the set of atomic formulas. The language $\mathcal{L}$ is generated by the following BNF:

$$\phi ::= p \mid \neg\phi \mid (\phi \wedge \phi) \mid (\phi \vee \phi)$$

for any $p \in P$.

We also deploy standard conventions for bracketing and definitions for other boolean connectives. Notably, $(\phi \rightarrow \psi)$ is defined as $(\neg\phi \vee \psi)$.

The *truthmaker model*, intuitively, decides a pair of sets—the set of verifiers and the set of falsifiers—of states in a state space for each atomic formula. Given a model we can recursively define the verification and falsification condition for any formula in $\mathcal{L}$. Formally:

Definition 1.5 (Modalized Set Space). A modalized set space is a triple $\langle S, S^\diamond, \sqsubseteq \rangle$ where $\langle S, \sqsubseteq \rangle$ is a set space and $S^\diamond$ (set of all possible states) is a non-empty subset of S such that $(s \in S^\diamond \& t \sqsubseteq$

$s) \implies t \in S^\diamond$.

Definition 1.6 (Model M). A model M is a quadruple $\langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$ where $\langle S, S^\diamond, \sqsubseteq \rangle$ is a modalized set space, and $|\cdot|$ is a valuation function which takes every $p \in P$ to a pair $\langle V, F \rangle$, where V and F are non-empty subsets of S that conform to the following conditions:

$$\begin{aligned} \text{Exclusivity: } & \forall v \in V \forall f \in F \ v \sqcup f \notin S^\diamond \\ \text{Exhaustivity: } & \forall s \in S^\diamond (\exists v \in V \ s \sqcup v \in S^\diamond \vee \exists f \in F \ s \sqcup f \in S^\diamond) \end{aligned}$$

Given $|p| = \langle V, F \rangle$, let $|p|^+$ denote the first component (i.e. V), and $|p|^-$ denote the second component (i.e. F).

Alternatively, we could allow V or F to be empty but never both. In fact, *Exhaustivity* already implies that it can never be the case that both V and F are empty, and *Exclusivity* grants that if one of them is empty, the other one would be $\{\square\}$.

Given a model M , we define a basic notion of verification and falsification, that is, *exact* verification $\Vdash$ and *exact* falsification $\dashv\vdash$:

Definition 1.7 (Exact Verification and Falsification for Formulas). Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$,

$$\begin{aligned} (\text{atom})^+ \ s \Vdash p & \iff s \in |p|^+ \\ (\text{atom})^- \ s \dashv\vdash p & \iff s \in |p|^- \\ (\text{negation})^+ \ s \Vdash \neg\phi & \iff s \dashv\vdash \phi \\ (\text{negation})^- \ s \dashv\vdash \neg\phi & \iff s \Vdash \phi \\ (\text{conjunction})^+ \ s \Vdash (\phi \wedge \psi) & \iff \exists t, u \text{ s.t. } t \Vdash \phi \ \& \ u \Vdash \psi \ \& \ s = t \sqcup u \\ (\text{conjunction})^- \ s \dashv\vdash (\phi \wedge \psi) & \iff s \dashv\vdash \phi \text{ or } s \dashv\vdash \psi \\ (\text{disjunction})^+ \ s \Vdash (\phi \vee \psi) & \iff s \Vdash \phi \text{ or } s \Vdash \psi \\ (\text{disjunction})^- \ s \dashv\vdash (\phi \vee \psi) & \iff \exists t, u \text{ s.t. } t \dashv\vdash \phi \ \& \ u \dashv\vdash \psi \ \& \ s = t \sqcup u \end{aligned}$$

Notice that the exact verifier of $\phi \wedge \psi$ does not always exactly verify ϕ . Likewise, the exact falsifier of $\phi \vee \psi$ does not always exactly falsify ϕ . Intuitively, a state s exactly verifies ϕ only if s is wholly relevant to the truth of ϕ . For example, the state that it is raining is wholly relevant to the truth of the statement “it is raining”, but the state that it is raining and there is a dog lying in the gutter is not wholly relevant to the truth of the statement “it is raining”; it is merely *partially* relevant. Similarly with the case of exact falsification.

Yet we could say the state that it is raining and there is a dog lying in the gutter *inexactly* verifies the statement “it is raining”. With the mereological structure of states in hand, we can define *inexact verification* $\Vdash\!\!\!\triangleright$ and *inexact falsification* $\triangleleft\!\!\!\Vdash$ as follows:

Definition 1.8 (Inexact Verification and Falsification for Formulas). Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$,

$$\begin{aligned} (\text{inexact})^+ \ s \Vdash\!\!\!\triangleright \phi & \iff \exists t \sqsubseteq s \text{ s.t. } t \Vdash \phi \\ (\text{inexact})^- \ s \triangleleft\!\!\!\Vdash \phi & \iff \exists t \sqsubseteq s \text{ s.t. } t \dashv\vdash \phi \end{aligned}$$

Finally, we define the notion of *subject matter*:

Definition 1.9 (Subject Matter). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$, let $\mathcal{V}(\phi)$ denote the set of all exact verifiers of ϕ ; and $\mathcal{F}(\phi)$ denote the set of all exact falsifiers of ϕ .*

The subject matter of ϕ $sm(\phi)$ is a state in S such that $sm(\phi) = \sqcup \mathcal{V}(\phi) \sqcup \sqcup \mathcal{F}(\phi)$.¹

2 Simple Frameworks

With the materials from truthmaker semantics, now I develop two simple frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$ that immediately arise from the idea that the expansion and contraction operate on a belief-state set $\mathcal{S}^\mathcal{K}$. The two simple frameworks will only differ in how to determine the belief set $\mathcal{K}$ from $\mathcal{S}^\mathcal{K}$. Thus the definitions in this section apply to both frameworks if not otherwise noted.

In simple frameworks we do not pose any constraint on $\mathcal{S}^\mathcal{K}$:

Definition 2.1 (Simple Belief-State Set). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$, a belief-state set $\mathcal{S}^\mathcal{K}$ is a subset of S .*

In simple frameworks, an expansion of $\mathcal{S}^\mathcal{K}$ with a state σ ($\mathcal{S}^\mathcal{K} + \sigma$) simply adds σ to $\mathcal{S}^\mathcal{K}$ if not already contained; and a contraction of $\mathcal{S}^\mathcal{K}$ with σ ($\mathcal{S}^\mathcal{K} \div \sigma$) simply extracts σ from $\mathcal{S}^\mathcal{K}$ only if it was originally contained. Thus:

Definition 2.2 (Expansion and Contraction in Simple Frameworks). *Given a belief-state set $\mathcal{S}^\mathcal{K}$, $\mathcal{S}^\mathcal{K} + \sigma = \mathcal{S}^\mathcal{K} \cup \{\sigma\}$; and $\mathcal{S}^\mathcal{K} \div \sigma = \mathcal{S}^\mathcal{K} \setminus \{\sigma\}$.*

As the ontology of states does not matter for our purpose, the ontology of belief-state sets are also not required. Yet intuitively, one can regard $\mathcal{S}^\mathcal{K}$ as a set of states that the modelled epistemic agent is epistemically committed to; that is, the set of sources of all its beliefs. My preferred understanding is to take $\mathcal{S}^\mathcal{K}$ as a model for the modelled epistemic agent's ‘representation’ of reality, in the broadest sense. Thus, the belief set would contain every formula that would be actually verified if the representation were totally correct. Since we have two different notions of verification: exact and inexact, we would have two different frameworks that decide $\mathcal{K}$ from $\mathcal{S}^\mathcal{K}$:

Definition 2.3 (Belief Set in $\mathcal{E}_0$ and $\mathcal{I}_0$). *In framework $\mathcal{E}_0$, given a simple belief-state set $\mathcal{S}^\mathcal{K}$, a belief set $\mathcal{K} = \{\phi : \exists \sigma \in \mathcal{S}^\mathcal{K} \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.*

In framework $\mathcal{I}_0$, given a simple belief-state set $\mathcal{S}^\mathcal{K}$, a belief set $\mathcal{K} = \{\phi : \exists \sigma \in \mathcal{S}^\mathcal{K} \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.

Notice that the two basic frameworks only differ in this respect. We could define a framework as an n -tuple. Thus:

Definition 2.4 (Frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$). *Given a model M , a framework $\mathcal{E}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_\mathcal{E} \rangle$ where Σ_0 is a set of belief-state sets $\{\mathcal{S}^\mathcal{K} : \mathcal{S}^\mathcal{K} \subseteq S\}$, $+$ and $\div$ are expansion and contraction respectively, and $\Delta_\mathcal{E}$ are decision functions from the set of belief-state sets to the set of belief sets such that $\Delta_\mathcal{E}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$.*

¹The standard definition of subject matter is in terms of *propositions* defined in terms of states: a *bilateral proposition* P is defined as $\langle P^+, P^- \rangle$ where P^+ and P^- contain its verifiers and falsifiers respectively. Then the subject matter of P is either the pair $\langle \sqcup P^+, \sqcup P^- \rangle$ (*differentiated subject matter*) or the fusion of $\sqcup P^+$ and $\sqcup P^-$ (*comprehensive subject matter*). I intentionally deviated from the standard definition for the project of this paper a commitment to the theory of truthmaker content is not required. The preference for the comprehensive over the differentiated subject matter in epistemic settings follows Jago (2024).

Given a model M , a framework $\mathcal{I}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{I}} \rangle$ where $\Delta_{\mathcal{I}}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi\} \subseteq \mathcal{L}$, and everything else is as defined above.

One might desire there to be additional notions of expansion and contraction that operate on $\mathcal{K}$. Intuitively, the expansion of $\mathcal{K}$ with a formula ϕ ($\mathcal{K} \oplus \phi$) would be the resulting belief set of the belief-state set $\mathcal{S}^{\mathcal{K}}$ expanded with *some* verifiers of ϕ ; the contraction of $\mathcal{K}$ with a formula ϕ ($\mathcal{K} \ominus \phi$) would be the resulting belief set of the belief-state set $\mathcal{S}^{\mathcal{K}}$ contracted with *all* verifiers of ϕ .

However, it should be noticed that given a definite $\mathcal{S}^{\mathcal{K}}$, the resulting belief set $\mathcal{K}$ is definite in both $\mathcal{E}_0$ and $\mathcal{I}_0$; whereas given a definite belief set one cannot track down to a unique belief-state set. So let us denote the particular belief set decided by a belief-state set $\mathcal{S}^{\mathcal{K}}$ as $D(\mathcal{S}^{\mathcal{K}})$. More specifically, the belief set decided by a belief-state set $\mathcal{S}^{\mathcal{K}}$ in a framework X will be denoted by $D_X(\mathcal{S}^{\mathcal{K}})$. The specification of the framework will be omitted unless context is insufficient to disambiguate. Then, the expansion and contraction for a belief set should be defined as follows:

Definition 2.5 (Operations on Belief Set in Simple Frameworks). *Given a belief-state set $\mathcal{S}^{\mathcal{K}}$ and a selection function γ ,*

$$(\mathcal{E}_0^{\oplus}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \oplus \phi = \gamma(D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}} + \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{E}_0^{\ominus}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{E}_0} \phi = D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}} \setminus \{\sigma : \sigma \Vdash \phi\})$$

$$(\mathcal{I}_0^{\oplus}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \oplus \phi = \gamma(D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}} + \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{I}_0^{\ominus}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{I}_0} \phi = D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}} \setminus \{\sigma : \sigma \Vdash \phi\})$$

The revision on a belief set ($\circledast$) is defined via Levi Identity:

$$(\mathcal{E}_0^{\circledast}) \ D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \circledast_{\mathcal{E}_0} \phi = (D_{\mathcal{E}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{E}_0} \neg\phi) \oplus \phi$$

$$(\mathcal{I}_0^{\circledast}) \ D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \circledast_{\mathcal{I}_0} \phi = (D_{\mathcal{I}_0}(\mathcal{S}^{\mathcal{K}}) \ominus_{\mathcal{I}_0} \neg\phi) \oplus \phi$$

This yields the result that expansion, and thus revision, on a belief set is still not definite in $\mathcal{E}_0$ nor in $\mathcal{I}_0$, for a formula can have multiple verifiers, exact or inexact. But I have an excuse for that. As I said, we can think of $\mathcal{S}^{\mathcal{K}}$ as the set of all doxastic sources, or reasons, the modelled epistemic agent has upon forming beliefs. So it seems natural to me that, given a belief-state set, one cannot determine what would be the result after one acquired a belief, for it is not specified what is the exact source of that acquired belief. However, the definition also allows the expansion to infringe *minimal action* for expansion (see Table 1). This feature, I admit, may be disputed. One might add the clause “if $\phi \in D(\mathcal{S}^{\mathcal{K}})$, $D(\mathcal{S}^{\mathcal{K}}) \oplus \phi = D(\mathcal{S}^{\mathcal{K}})$ ” to the definition to retain the minimal action postulate, if one wishes. On the contrary, it seems that, given a belief-state set, we could determine what would be the result after one discarded a belief: the agent would have lost *all* the sources for the discarded belief. Also notice that in case of contraction, *minimal action* for contraction (see Table 2) holds.

Now let us observe how the two simple frameworks satisfy every desideratum except **Local Rationality**.

(D1) Definite Contraction

It would be easy to check that the contraction (on a belief-state set) is uniquely determined, since the relative complement is uniquely determined.

One interesting feature of these frameworks is that we can now define the entrenchment of

the belief that $\phi \in D(\mathcal{S}^\mathcal{K})$ by inspecting $\mathcal{S}^\mathcal{K}$. As for verification, we have exact and inexact entrenchment, such that:

Definition 2.6 ((Entrenchment)). *Given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$,*

The exact entrenchment of ϕ is the number of exact verifiers of ϕ in $\mathcal{S}^\mathcal{K}$, i.e., $\text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$

The inexact entrenchment of ϕ is the number of inexact verifiers of ϕ in $\mathcal{S}^\mathcal{K}$, i.e., $\text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$

The intuitive idea behind is that the modelled epistemic agent is more entrenched to the belief that ϕ just in case it has more sources or reasons for believing that ϕ . Observe that different frameworks get along well with different notions of entrenchment: the natural notion of entrenchment in framework $\mathcal{E}_0$ is *exact entrenchment*, while in framework $\mathcal{I}_0$ it is *inexact entrenchment*.

With the notion of entrenchment, one thing we can now make sense of is the phenomenon that sometimes expanding or contracting on $\mathcal{S}^\mathcal{K}$, which makes a difference to the belief-state set, may still not result in any difference to $\mathcal{K}$. Suppose the state $t \notin \mathcal{S}^\mathcal{K}$ only verifies ϕ , but $s \in \mathcal{S}^\mathcal{K}$ such that $s \Vdash \phi$, and all fusions of t and any state in $\mathcal{S}^\mathcal{K}$ are not in $\mathcal{S}^\mathcal{K}$. Then $\mathcal{S}^\mathcal{K} + t$ would make no difference in $\mathcal{K}$, under both frameworks $\mathcal{E}_0$ and $\mathcal{I}_0$; likewise $\mathcal{S}^\mathcal{K} \cup \{t\} \div t$ would make no difference in $\mathcal{K}$. This, I submit, can be made sense of by the notion of entrenchment we just defined. Although the belief set remains the same, by expanding with t the entrenchment of ϕ has increased, i.e., the belief that ϕ is reinforced. Likewise, by contracting with t the entrenchment of ϕ has decreased, i.e., the belief that ϕ is weakened.

(D2) Logical Ignorance

This is relatively easy to show. In fact, it is rather difficult to construct a belief-state set that makes all logical truths contained in the belief set. To establish logical omniscience, the belief-state set must include, for any formula, either the verifier of it or the falsifier of it. Otherwise there are some logical truths that are not members of $\mathcal{K}$. One extreme example is where $\mathcal{S}^\mathcal{K} = \emptyset$, which is allowed under the definition. Then by the definition of a belief set, $\mathcal{K}$ would contain no formula at all, needless to say all logically true formulas.

(D3) Paraconsistency

The fact that the frameworks satisfy **Paraconsistency** is also very direct. Notice that in simple frameworks there is no constraint on $\mathcal{S}^\mathcal{K}$ that it should not contain both verifiers and falsifiers for some formulas. Thus, suppose $\mathcal{S}^\mathcal{K} = \{s, t\}$ where s verifies (exactly for $\mathcal{E}_0$, inexactly for $\mathcal{I}_0$. Similarly, for below) ϕ and t falsifies ϕ (therefore verifies $\neg\phi$); and each state in $\mathcal{S}^\mathcal{K}$ verifies or falsifies nothing else. Then $D(\mathcal{S}^\mathcal{K}) = \{\phi, \neg\phi\}$, without introducing all formulas in $\mathcal{L}$ into $D(\mathcal{S}^\mathcal{K})$.

However, this may also be thought of as a weakness of simple frameworks, for it means that the frameworks allow blatant contradictions. The details will be discussed below when discussing **Local Rationality**.

(D5) Hyperintensionality

Hyperintensionality is automatically granted in our frameworks, for the underlying truthmaker model is already hyperintensional. Suppose $s_1 \Vdash p$ and $s_2 \dashv\vdash p$; $t_1 \Vdash q$ and $t_2 \dashv\vdash q$. What follows

is that although $\vdash p \vee \neg p$ and $\vdash q \vee \neg q$, they may have different verifiers: for $s_1 \Vdash p \vee \neg p$ and $t_1 \Vdash q \vee \neg q$. Likewise, although $p \wedge \neg p$ and $q \wedge \neg q$ are both logical falsities, they may have different verifiers: for $s_1 \sqcup s_2 \Vdash p \wedge \neg p$ and $t_1 \sqcup t_2 \Vdash q \wedge \neg q$. Recall that there were no constraints on the model M that any states must verify logical truths or falsify logical falsities. Thus expanding or contracting with $p \vee \neg p$ on $\mathcal{K}$ is not equivalent to expanding or contracting with $q \vee \neg q$; likewise expanding or contracting with $p \wedge \neg p$ on $\mathcal{K}$ is not equivalent to expanding or contracting with $q \wedge \neg q$.

(D6) Pertinent Recovery

Recall the problematic case of recovery: ϕ strictly implies ψ , so contracting with ψ would result in removing ϕ from $\mathcal{K}$; yet by *recovery* (in Table 2) expanding with ψ would recover not only ψ but also ϕ . However, this contradicts our daily observations, for the modelled epistemic agent could believe ψ with some sources that do not force it to believe ϕ .

The proposed frameworks deal with this kind of problem particularly well. Let us suppose all verifiers of ϕ , including s , are also verifiers of ψ , but there is a state t which verifies ψ but not ϕ . By supposition, if $s \in \mathcal{S}^\mathcal{K}$, $\mathcal{K}$ will include both ϕ and ψ ; and $\{\phi, \psi\} \cap (D(\mathcal{S}^\mathcal{K}) \ominus \psi) = \emptyset$. But still, it is not necessarily the case that $\phi \in (D(\mathcal{S}^\mathcal{K}) \ominus \psi) \oplus \psi$. Since expansion on a belief set is not uniquely determined, $(D(\mathcal{S}^\mathcal{K}) \ominus \psi) \oplus \psi$ might be the set that contains the state s that verifies both ϕ and ψ , but might as well be the set that contains the state t that does not verify ϕ , and thus $\phi \notin (D(\mathcal{S}^\mathcal{K}) \ominus \psi) \oplus \psi$.

So the idea that, given merely the assumptions that ϕ entails ψ and $\{\psi, \phi\} \subseteq \mathcal{K}$, we cannot know whether $(\mathcal{K} \ominus \psi) \oplus \psi$ includes ϕ is preserved. We can, of course, know whether s is in $(\mathcal{S}^\mathcal{K} \div t) + t$ given information about whether $\{s\} \subseteq \mathcal{S}^\mathcal{K}$. But as noted in §1.1.4, the recovery formulated in terms of the source of belief—states and belief-state sets—is extremely compelling.

To sum up, the two simple frameworks - $\mathcal{E}_0$ and $\mathcal{I}_0$ already do very well in satisfying all the desiderata except **Local Rationality**. Yet they are untenable for they fail to satisfy the only remaining desideratum so drastically.

2.1 Rationality

As mentioned above, because there were no constraints on $\mathcal{S}^\mathcal{K}$ in both simple frameworks, they allow $\{\phi, \neg\phi\} \subseteq \mathcal{K}$, and also $\phi \wedge \neg\phi \in \mathcal{K}$ without any tension. Although I do not submit that the modelled agent should never be capable of having an inconsistent belief set, I do hold that an inconsistent belief set must cause some sort of ‘tension’. This is required if one wants to have a suitable notion of revision on $\mathcal{S}^\mathcal{K}$.

Recall that in simple frameworks revision is defined only on a belief set. This is because without any constraints on the consistency of $\mathcal{S}^\mathcal{K}$, any attempts to define revision would be arbitrary, or too strong to satisfy other desiderata. For instance, one might wish to define revising with σ on $\mathcal{S}^\mathcal{K}$ ($\mathcal{S}^\mathcal{K} * \sigma$) to be $(\mathcal{S}^\mathcal{K} \setminus \{\tau : \tau \sqcup \sigma \notin \mathcal{S}^\diamond\}) \cup \{\sigma\}$. The idea is that, if we were to revise with σ , we first contract with all states that are incompatible with σ , and then expand with σ . Then by definition of a model, $D(\mathcal{S}^\mathcal{K} * \sigma)$ will exclude all the formulas that are exactly falsified by σ . So far so good, but this also excludes all formulas in $D(\mathcal{S}^\mathcal{K})$ such that every verifier of it is an impossible state; i.e., a logical falsity, for $\tau \sqcup \sigma \notin \mathcal{S}^\diamond$ if $\tau \notin \mathcal{S}^\diamond$. So after a revision on a belief-state set with σ it would result in removing all logical falsities. This would infringe our desideratum on **Hyperintensionality**. What if σ is already an impossible state?

Actually the result is much more devastating: if $\sigma \notin S^\diamond$, according to the proposed definition of revision, $S^\mathcal{K} * \sigma = \{\sigma\}$, for any τ will fuse with σ to an impossible state. Since we wanted the modelled agent to have minimal changes even if it is forced to believe inconsistently, we might want to say this violates **Paraconsistency**.

Another unfortunate feature of simple frameworks is that *every* belief must be incorporated into the agent manually. $\mathcal{I}_0$ is perhaps better in this respect, for by believing a conjunction it forces the agent to believe each of its conjuncts, while $\mathcal{E}_0$ does not yield this feature since the exact verifier of a conjunction is not generally the exact verifier of its conjuncts. But it should be examined whether this feature is really desirable. I postpone this discussion to §2.3.

One especially problematic consequence of both basic frameworks is that one can believe that ϕ and $\phi \rightarrow \psi$ ($\iff \neg\phi \vee \psi$, since $\rightarrow$ is not a primitive operator in our language), yet not believe ψ . For the reason I will suggest in the next subsection, I admit there can be cases where one believes both $\phi \wedge \chi$ and $\phi \rightarrow \psi$, but not ψ . But it seems to me that a locally rational epistemic agent must believe that ψ , given that the agent believes that ϕ , and it also believes that either it is the case that $\neg\phi$ or ψ . In short, we want the modelled epistemic agent to be rational enough to have a belief set closed under modus ponens.

To sum up, simple frameworks fail to satisfy the ‘rationality’ part of **Local Rationality**, for it permits blatant inconsistency in $\mathcal{K}$, and also because it fails to make the belief set closed under modus ponens (of material conditional). One might additionally complain that it is not closed under conjunction elimination. It is, in some respects, too weak to satisfy the desideratum. However, it is sometimes too strong to satisfy the ‘locality’ part of **Local Rationality**.

2.2 Locality

In some sense, it may be said that to model the locality of the rationality of the modelled epistemic agent is more important to our present purpose. For, if one only wanted the belief set to be gently consistent, or closed under modus ponens or conjunction elimination, the traditional AGM framework is already sufficient. In the previous section, we observed some cases where the frameworks fail to yield even local rationality. But there are also cases where the frameworks yield consequences other than what we might desire.

One crucial failure of locality is that in both simple frameworks, the belief set is closed under disjunction introduction: if $\phi \in \mathcal{K}$, for any formula ψ , $\phi \vee \psi \in \mathcal{K}$. It is easy to notice that we do not believe this way: although Frege would have believed that Wilhelm II is the King of Prussia, he would not have believed that Wilhelm II is the King of Prussia or Friedrich Merz is the chancellor of Germany.

Besides intuitive implausibility, the closure under disjunction introduction also has further problems. For one thing, this makes any modelled epistemic agent believe *about* everything. Let us define the subject matter of a *set* of formulas X as the least upper bound of the set of all the subject matters of the formulas in X , formally:

Definition 2.7 (Subject Matter for Sets). *Given a model $M = \langle S, S^\diamond, \sqsubseteq, |\cdot| \rangle$, the subject matter of a set $X \subseteq \mathcal{L}$ is defined as $sm(X) = \bigsqcup \{\sigma : \exists \phi \in X \text{ s.t. } sm(\phi) = \sigma\}$.*

With the notion of subject matter for sets, we might want to investigate what the modelled epistemic agent believes *about*. One intriguing example is that we can now define a stronger notion of ignorance: the agent is ignorant about ϕ if $\{\phi, \neg\phi\} \cap \mathcal{K} = \emptyset$; but the agent is strongly

ignorant about ϕ if the only common part of $sm(\phi)$ and $sm(\mathcal{K})$ is $\square$.

Yet if $\mathcal{K}$ is closed under disjunction introduction, there are no formulas about which the agent is strongly ignorant. In fact, for any possible $\mathcal{K}$ that is not empty, $sm(\mathcal{K}) = sm(\mathcal{L})$ under both simple frameworks. This is clearly not desirable. It seems that until I searched on the internet to write this paragraph, I was strongly ignorant that Amr ibn al-Layth was an Amir of the Saffarid dynasty. But simple frameworks deny this intuition, for, since I believe that I have a hand, my belief-state set would include a state s that verifies the statement “I have a hand”, which also verifies “I have a hand or Amr ibn al-Layth was an Amir of the Saffarid dynasty”. And since the subject matter of the statement “Amr ibn al-Layth was an Amir of the Saffarid dynasty” is a part of the subject matter of the disjunction, I would not be strongly ignorant about the statement about Amr ibn al-Layth, which I have never heard before.

Thus, it would be better to have a belief set that is not closed under disjunction introduction. However, excluding the disjunctive consequence of beliefs should be carefully executed; for we do not wish to ban the modelled agent from having disjunctions as beliefs. Notice that any verifiers, exact or inexact, of a disjunction $\phi \vee \psi$ must be a verifier, exact or inexact, of at least one of its disjuncts ϕ or ψ . So if we restrict the modelled epistemic agent to believe $\phi \vee \psi$ on the basis of the belief that ϕ , the result is that none of the members of $\mathcal{K}$ is a disjunction. This suggests that we may need a more fine-grained constraint on locality.

2.3 Closure Under Conjunction Elimination?

Finally, let me also note a problem about conjunction. Notice that in $\mathcal{E}_0$ even if $\phi \wedge \psi \in \mathcal{K}$, it does not grant that $\{\phi, \psi\} \subseteq \mathcal{K}$, whereas in $\mathcal{I}_0$ it is guaranteed. So at first sight, it may seem as a reason to favor $\mathcal{I}_0$ over $\mathcal{E}_0$. One might therefore conclude that $\mathcal{K}$ should be defined from $\mathcal{S}^{\mathcal{K}}$ via the notion of *inexact* verification rather than the notion of *exact* verification.

The idea that the belief set should be closed under conjunction elimination seems plausible. However, I do not wish to adjudicate which notion of verification is suitable for defining $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$. The worry of using the exact notion of verification is as stated: the belief set can contain a conjunction without containing any of its conjuncts, which might make the modelled agent seem less rational than we wish it to be. Indeed, there might be something wrong with an agent if it believes $\phi \wedge \psi$ without believing that ϕ or ψ .

But using the inexact notion of verification is not without problems. We say ϕ is a conjunctive part of ψ (and ψ is a conjunctive whole of ϕ) iff: (a) for all $s \Vdash \psi$, there is a t such that $t \sqsubseteq s$ and $t \Vdash \phi$, and (b) for all $t \Vdash \phi$, there is an s such that $t \sqsubseteq s$ and $s \Vdash \psi$ (and (c) any falsifier of ϕ is a falsifier of ψ . Clause (c) may sometimes be omitted)². Notice that the definition of conjunctive part does not involve the syntactic structure of ϕ or ψ . Now, under $\mathcal{I}_0$ not only each conjunct of a conjunction in $\mathcal{K}$ is included in $\mathcal{K}$, but all conjunctive parts of a member of $\mathcal{K}$ are included in $\mathcal{K}$. This follows directly from the fact that any inexact verifier of a conjunctive whole would be an inexact verifier of its conjunctive part.

But it is unclear why the modelled epistemic agent should believe all the conjunctive parts of its beliefs, especially when the conjunctive part in question is not syntactically related to its whole. At least in some cases, it seems there is good reason to believe that believing the conjunctive whole gives sufficient condition to believe its conjunctive part regardless of their

²Again, the notion of conjunctive part is also usually defined in terms of propositions; but let us operate with a watered-down notion of conjunctive part of formulas.

syntax. For example, it is reasonable to think “ x is red” is a conjunctive part of “ x is scarlet”; and if one believes that x is scarlet, one has sufficient reason to believe that x is red (Jago, 2024).

But it should also be noticed that sometimes the notion of conjunctive parthood is used to formulate semantic or metaphysical notions. It is customary to think that if ϕ is a conjunctive part of ψ , the proposition that ϕ is a *partial content* of the proposition that ψ (Fine 2016, 2017a, 2017b). And it is also suggested that ϕ is a strict partial ground of ψ iff ϕ is a conjunctive part of ψ but not vice versa (Fine, 2012); or that ϕ is a partial essence of ψ iff ϕ is a conjunctive part of ψ (Correia & Skiles, 2017). If these suggestions are correct, defining $\mathcal{K}$ to contain every formula inexactly verified by some members of $\mathcal{S}^{\mathcal{K}}$ would result in the modelled agent believing every partial content, every strict partial ground, and every partial essence of its beliefs—which sounds very suspicious. Of course, none of these suggestions are unchallenged; but $\mathcal{I}_0$ or its kin would be just as suspicious as those accounts are compelling.

Still, I personally favor defining $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$ by inexact verification. At least the semantic application of conjunctive parthood does not seem to threaten $\mathcal{I}_0$, for it may be reasonably claimed that an ideal epistemic agent should not believe ϕ if it is not fully aware of its semantic content. Nevertheless, the consideration above shows that frameworks that define $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$ by exact verification also deserve substantial interest. Also, as we will see in the next section, the approach validates more fine-grained distinctions among the constraints on $\mathcal{K}$. For such reasons both approaches will be developed in the remainder of the paper.

A final remark. There is in fact a moderate alternative between exact and inexact approaches: given a model M and a belief-state set $\mathcal{S}^{\mathcal{K}}$, we first define $\mathcal{K}_0$ to be the set of all formulas *exactly* verified by some members of $\mathcal{S}^{\mathcal{K}}$; $\mathcal{K}_{n+1}$ is defined as $\mathcal{K}_n \cup \{\phi, \psi : \phi \wedge \psi \in \mathcal{K}_n\}$; and finally the belief set $\mathcal{K}$ is defined as $\bigcup_{n=0}^{\infty} \mathcal{K}_n$. This yields the result that $\mathcal{K}$ is closed under conjunction elimination, but not necessarily closed under conjunctive parthood. I will not develop this kind of approach further in this paper, but this is indeed an option.

3 More Frameworks!

3.1 The Locality Constraint

Let us solve the problems of simple frameworks step-by-step. It turns out that the problem of disjunction is much easier to deal with. We only modify the definition of the belief set to get the simple frameworks supplemented with the locality constraint:

Definition 3.1 (Frameworks $\mathcal{EL}_0$). *Given a model M , the framework $\mathcal{EL}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{EL}} \rangle$ where $\Delta_{\mathcal{EL}}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi \ \& \ \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \Box(\tau \sqsubseteq \text{sm}(\psi) \ \& \ \tau \sqsubseteq \Box X)\} \subseteq \mathcal{L}$, and everything else is as defined in **Definition 2.4**.*

Definition 3.2 (Frameworks $\mathcal{IL}_0$). *Given a model M , the framework $\mathcal{IL}_0$ is a quintuple $\langle M, \Sigma_0, +, \div, \Delta_{\mathcal{IL}} \rangle$ where $\Delta_{\mathcal{IL}}(X) = \{\phi : \exists \sigma \in X \text{ s.t. } \sigma \Vdash \phi \ \& \ \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \Box(\tau \sqsubseteq \text{sm}(\psi) \ \& \ \tau \sqsubseteq \Box X)\} \subseteq \mathcal{L}$, and everything else is as defined in **Definition 2.4**.*

The expansion and contraction with formulas under the framework $\mathcal{EL}_0$ are defined as they were defined under $\mathcal{E}_0$; and similarly under $\mathcal{IL}_0$ with $\mathcal{I}_0$.

The only difference in these two new frameworks is that we have an additional locality constraint on deciding $\mathcal{K}$ from $\mathcal{S}^{\mathcal{K}}$: intuitively, the subject matter of formulas in $\mathcal{K}$ must have something in common with $\mathcal{S}^{\mathcal{K}}$. Given $\phi \in \mathcal{K}$, the locality constraint yields that $\phi \vee \psi \in \mathcal{K}$

if and only if $sm(\psi)$ has a non-trivial common part with $\sqcup \mathcal{S}^{\mathcal{K}}$. Roughly, by incorporating the locality constraint on Δ , we allow the modelled epistemic agent to have disjunctive belief, but only to the extent that each of the disjuncts is “partially entertained in the representation”, positively or negatively.

This restores the intuition that epistemic agents can be strongly ignorant of some formulas. Suppose $\mathcal{S}^{\mathcal{K}} = \{s_1, s_2\}$, $|p|^+ = \{t_1\}$ and $|p|^- = \{t_2\}$, and $s_1 \Vdash q$ and $s_2 \nVdash r$. Finally, suppose $sm(p)(= t_1 \sqcup t_2) \not\sqsubseteq \sqcup \mathcal{S}^{\mathcal{K}} (= s_1 \sqcup s_2)$. Then, under both frameworks $\mathcal{EL}_0$ and $\mathcal{IL}_0$, $\{q \vee r, q \vee \neg r\} \subseteq \mathcal{K}$, but $\{q \vee p, q \vee \neg p\} \cap \mathcal{K} = \emptyset$. Furthermore, if neither q nor r has exact verifiers or falsifiers which have t_1 or t_2 as part, and there are no other atomic formulas, we could conclude that the agent which has such a belief set is strongly ignorant about p , i.e., the only common part of $sm(p)$ and $sm(\mathcal{K})$ is $\square$.

3.2 Consistency Constraints

By contrast, the problem of rationality is difficult—not because it requires a highly complex formal mechanism to solve, but because there are so many different degrees of rationality one might want to impose on the modelled epistemic agent.

One main problem of simple frameworks addressed in §2.1 was that simple frameworks allow the belief set to consist of both ϕ and $\neg\phi$. However, if a given $\mathcal{S}^{\mathcal{K}}$ already consists of both a verifier and a falsifier of a formula ϕ (thus verifiers for ϕ and $\neg\phi$), it would be arbitrary to choose one of ϕ or $\neg\phi$ to include in $\mathcal{K}$. To use the metaphor of “source”: if an epistemic agent has both sources, or reasons, for believing that ϕ and believing that $\neg\phi$, it would seem incorrect to ascribe only one of the beliefs to the agent. Instead, since we want to model a somewhat ideal epistemic agent, not all sorts of bizarre believers, we could simply forbid the modelled epistemic agent from having both a verifier and a falsifier for any formula ϕ in its “source”.

This idea immediately suggests some constraints we can impose on the definition of $\mathcal{S}^{\mathcal{K}}$. Since we have two notions of verification and falsification, there is, in fact, more than one constraint that arises from the idea above.

Definition 3.3 (Set of Belief-State Sets with Consistency Constraints). *Given a model M ,*

- (0) $\Sigma_0 = \{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S\}$
- (1) $\Sigma_1 = \{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^{\mathcal{K}}, \neg \exists \phi (s \Vdash \phi \ \& \ t \nVdash \phi)\}$
- (2a) $\Sigma_{2a} = \{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^{\mathcal{K}}, \neg \exists \phi (s \Vdash \phi \ \& \ t \lhd \phi)\}$
- (2b) $\Sigma_{2b} = \{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^{\mathcal{K}}, \neg \exists \phi (s \Vdash \phi \ \& \ t \nVdash \phi)\}$
- (3) $\Sigma_3 = \Sigma_{2a} \cap \Sigma_{2b}$
- (4) $\Sigma_4 = \{\mathcal{S}^{\mathcal{K}} : \mathcal{S}^{\mathcal{K}} \subseteq S \ \& \ \forall s, t \in \mathcal{S}^{\mathcal{K}}, \neg \exists \phi (s \Vdash \phi \ \& \ t \lhd \phi)\}$

It would be easy to check that $\Sigma_4 \subseteq \Sigma_3 \subseteq \Sigma_{2a} \subseteq \Sigma_1 \subseteq \Sigma_0$, and $\Sigma_4 \subseteq \Sigma_3 \subseteq \Sigma_{2b} \subseteq \Sigma_1 \subseteq \Sigma_0$.

Before we proceed to define our new frameworks utilizing consistency constraints, we need to first refine our definition of $+$. If we continue to regard $\mathcal{S}^{\mathcal{K}} + \sigma = \mathcal{S}^{\mathcal{K}} \cup \{\sigma\}$, the result of expansion would not generally be a belief-state set, i.e., a member of Σ_n , since $\mathcal{S}^{\mathcal{K}}$ may contain a falsifier of a formula that σ verifies (or vice versa). The modification, however, is rather

simple, for we only need to add a condition for an expansion operation to succeed, as there was a condition for a contraction operation to succeed in the original AGM framework.

Definition 3.4 (Expansion and Contraction in General). *Given a belief-state set $\mathcal{S}^\mathcal{K} \in \Sigma_n \subseteq \Sigma_0$ and $\sigma \in S$,*

$$\mathcal{S}^\mathcal{K} +_n \sigma = \begin{cases} \mathcal{S}^\mathcal{K} \cup \{\sigma\} & \text{if } \mathcal{S}^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ \mathcal{S}^\mathcal{K} & \text{otherwise} \end{cases}$$

$$\mathcal{S}^\mathcal{K} \div \sigma = \mathcal{S}^\mathcal{K} \setminus \{\sigma\}$$

With the new conception of expansion with states, we can now further define a suitable notion of revision with states. The idea is that, if one revises with σ on $\mathcal{S}^\mathcal{K}$, we first contract a minimal number of states from $\mathcal{S}^\mathcal{K}$ to make the expansion with σ successful, and then expand with σ using the result of the contractions. First, we define conflict relations Θ that correspond to each consistency constraint:

Definition 3.5 (Conflict Relations Corresponding to Consistency Constraints). *Given a model M and σ, τ both in S ,*

- (0) $\forall \sigma, \tau \neg \Theta_0(\sigma, \tau)$
- (1) $\Theta_1(\sigma, \tau) \iff \exists \phi ((\sigma \Vdash \phi \ \& \ \tau \dashv \Vdash \phi) \vee (\tau \Vdash \phi \ \& \ \sigma \dashv \Vdash \phi))$
- (2a) $\Theta_{2a}(\sigma, \tau) \iff \exists \phi ((\sigma \Vdash \phi \ \& \ \tau \lessdot \Vdash \phi) \vee (\tau \Vdash \phi \ \& \ \sigma \lessdot \Vdash \phi))$
- (2b) $\Theta_{2b}(\sigma, \tau) \iff \exists \phi ((\sigma \gg \Vdash \phi \ \& \ \tau \dashv \Vdash \phi) \vee (\tau \gg \Vdash \phi \ \& \ \sigma \dashv \Vdash \phi))$
- (3) $\Theta_3(\sigma, \tau) \iff \Theta_{2a}(\sigma, \tau) \vee \Theta_{2b}(\sigma, \tau)$
- (4) $\Theta_4(\sigma, \tau) \iff \exists \phi ((\sigma \gg \Vdash \phi \ \& \ \tau \lessdot \Vdash \phi) \vee (\tau \gg \Vdash \phi \ \& \ \sigma \lessdot \Vdash \phi))$

Then we can define revision with states via Quasi-Levi-Identity:

Definition 3.6 (Revision with Consistency Constraint). *Given a model M and $\sigma \in S$*

$$\mathcal{S}^\mathcal{K} *_n \sigma = (\mathcal{S}^\mathcal{K} \setminus \{\tau : \Theta_n(\sigma, \tau)\}) \cup \{\sigma\}.$$

That is, let $\langle \tau_i \rangle_{i=1}^k$ be an arbitrary enumeration of all states in $\mathcal{S}^\mathcal{K}$ such that $\Theta_n(\sigma, \tau_i)$ holds. We construct a sequence of contractions where $S_0 = \mathcal{S}^\mathcal{K}$ and $S_{i+1} = S_i \div \tau_i$. The revision operation is then defined purely in terms of contraction and expansion as:

$$\mathcal{S}^\mathcal{K} *_n \sigma = S_k +_n \sigma$$

Now we are ready to define new frameworks using consistency constraints. In principle, we could compose numerous frameworks with the resources given, but only four of them would be worth closer examination:

Definition 3.7 (Frameworks $\mathcal{EL}_1$, $\mathcal{EL}_3$, $\mathcal{EL}_4$, and $\mathcal{IL}_4$). *Given a model M ,*

The framework $\mathcal{EL}_1$ is a quintuple $\langle M, \Sigma_1, +_1, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{EL}_3$ is a quintuple $\langle M, \Sigma_3, +_3, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{EL}_4$ is a quintuple $\langle M, \Sigma_4, +_4, \div, \Delta_{\mathcal{EL}} \rangle$;

The framework $\mathcal{IL}_4$ is a quintuple $\langle M, \Sigma_4, +_4, \div, \Delta_{\mathcal{IL}} \rangle$.

These (perhaps except $\mathcal{EL}_1$) are the first frameworks presented in the paper which satisfy all the desiderata in §1.1. Notice that any $\mathcal{IL}$ -series framework which has a weaker consistency constraint than that of Σ_4 would still suffer from blatant inconsistency ($\{\phi, \neg\phi\} \subseteq \mathcal{K}$). Suppose $s \Vdash p$ and $t \Vdash \neg p$, but p is not exactly verified or falsified by any of them. Furthermore, suppose s and t do not exactly verify or falsify any formula. Then under a consistency constraint weaker than that of Σ_4 it could permit $\{s, t\} \subseteq \mathcal{S}^{\mathcal{K}}$, and by the definition of $\Delta_{\mathcal{IL}}$, it could therefore permit $\{p, \neg p\} \subseteq \mathcal{K}$. However, obviously, $\mathcal{EL}_1$ would not suffer from the same problem.

Interestingly, the consistency constraint can also guarantee the belief set to be closed under modus ponens. For convenience, I will only show that $\mathcal{EL}_1$ satisfies this desideratum, and it will be obvious that the other three frameworks also have this feature. We assume the contrary, that $\{\phi, \neg\phi \vee \psi\} \subseteq \mathcal{K}$ but $\psi \notin \mathcal{K}$. This implies that $\mathcal{S}^{\mathcal{K}}$ contains some exact verifier of ϕ and does not contain any exact verifier of ψ ; and by the consistency constraint it follows that $\mathcal{S}^{\mathcal{K}}$ does not contain any exact falsifier of ϕ . However, since any exact verifier of $\neg\phi \vee \psi$ is either an exact falsifier of ϕ or an exact verifier of ψ , and neither of these is in $\mathcal{S}^{\mathcal{K}}$, $\neg\phi \vee \psi \notin \mathcal{K}$ —a contradiction. Therefore, for any formulas ϕ and ψ , if $\{\phi, \neg\phi \vee \psi\} \subseteq \mathcal{K}$ then $\psi \in \mathcal{K}$, i.e., $\mathcal{K}$ is closed under modus ponens.

The difference between the three new $\mathcal{EL}$ -series frameworks can be appreciated by examining which sets of formulas are possible subsets of $\mathcal{K}$ under different frameworks. For example, we can observe that if one says $\mathcal{EL}_1$ satisfies Local Rationality, one is saying so in a quite weak sense of rationality, since it is possible that $\{p \wedge \neg p\} \subseteq \mathcal{K}$, or $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$. Suppose s_1 and s_2 are the sole exact verifier and falsifier of p respectively, and t_1 and t_2 are the sole exact verifier and falsifier of q respectively. Let us further suppose no mentioned state is a part of another mentioned state, and p and q are the only atomic formulas. Then a necessary and sufficient condition for $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$ to be the case under $\mathcal{EL}_1$ is that $\{s_1 \sqcup t_1, s_2 \sqcup t_2\} \subseteq \mathcal{S}^{\mathcal{K}}$. It should be noticed that there are no formulas *exactly* verified by $s_1 \sqcup t_1$ yet *exactly* falsified by $s_2 \sqcup t_2$, and also vice versa: atomic formulas p or q (and therefore $\neg p$ or $\neg q$) are merely inexactly verified/falsified by them; $p \vee q$ (and therefore $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$) is exactly falsified (verified) by $s_2 \sqcup t_2$ yet merely inexactly verified (falsified) by $s_1 \sqcup t_1$; and $p \wedge q$ (and therefore $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$) is exactly verified (falsified) by $s_1 \sqcup t_1$ yet merely inexactly falsified (verified) by $s_2 \sqcup t_2$.

$\{p \wedge q, \neg p \wedge \neg q\}$, however, is not a possible subset of $\mathcal{K}$ under $\mathcal{EL}_3$, and *a fortiori* $\mathcal{EL}_4$. For any two states σ and τ , if $\sigma \Vdash p \wedge q$ and $\tau \Vdash \neg p \wedge \neg q$, then τ would have an exact verifier of $\neg p$ as part (possibly improper), which is an exact falsifier of p , and therefore an exact falsifier of $p \wedge q$. Thus τ would have an exact falsifier of $p \wedge q$ as part, i.e., is an inexact falsifier of $p \wedge q$. Therefore, any such set $\{\sigma, \tau\} \not\subseteq \mathcal{S}^{\mathcal{K}}$ under $\mathcal{EL}_3$ and $\mathcal{EL}_4$, and therefore $\{p \wedge q, \neg p \wedge \neg q\} \subseteq \mathcal{K}$ is also impossible under those frameworks.

Likewise, we might wish to find a set of formulas that is possibly a subset of $\mathcal{K}$ under $\mathcal{EL}_3$ but not under $\mathcal{EL}_4$. This is, however, impossible. In fact, any possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$, and vice versa. A set is a possible $\mathcal{K}$ -subset under a framework $\mathcal{X}_n$ if and only if there is a model M giving $\mathcal{X}_n$ such that there is a $\mathcal{S}^{\mathcal{K}} \in \Sigma_n$ where the set is a subset of $\Delta_X(\mathcal{S}^{\mathcal{K}})$. Then:

Theorem 3.1 (Equivalence of Possible $\mathcal{K}$ -subset Under $\mathcal{EL}_3$ and $\mathcal{EL}_4$). (a) *Every possible $\mathcal{K}$ -*

subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$, and (b) every possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$.

Proof for (a): We prove the contrapositive. Since Γ is not a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$, by definition of $\Delta_{\mathcal{EL}}$, there are ϕ and ψ in Γ such that $\forall \sigma, \tau ((\sigma \Vdash \phi \ \& \ \tau \Vdash \psi) \implies \Theta_4(\sigma, \tau))$, that is, $\exists \chi ((\sigma \Vdash \chi \ \& \ \tau \Vdash \chi) \vee (\tau \Vdash \chi \ \& \ \sigma \Vdash \chi))$. For the sake of simplicity, suppose s and t are the sole exact verifiers for ϕ and ψ respectively, and χ_1 is the witness of χ such that $s \Vdash \chi_1 \ \& \ t \Vdash \chi_1$. Then, there exist $s' \sqsubseteq s$ and $t' \sqsubseteq t$ such that $s' \Vdash \chi_1$ and $t' \nVdash \chi_1$; which further yields $t' \nVdash \chi_1 \wedge \phi$, and therefore $t \nVdash \chi_1 \wedge \phi$, and $s' \sqcup s \Vdash \chi_1 \wedge \phi$. Now, since $s' \sqsubseteq s$, we get $s' \sqcup s = s$, and therefore $s \Vdash \chi_1 \wedge \phi$. By the definition of $\Delta_{\mathcal{EL}}$ and the supposition that s and t are the unique exact verifiers of ϕ and ψ respectively, Γ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ if and only if $\neg \Theta_3(s, t)$. But since $s \Vdash \chi_1 \wedge \phi$ and $t \nVdash \chi_1 \wedge \phi$, $\Theta_{2a}(s, t)$ and therefore $\Theta_3(s, t)$. Now suppose χ_2 is the witness of χ such that $s \nVdash \chi_2$ and $t \Vdash \chi_2$. By similar reasoning, we can get the result that $t \Vdash \chi_2 \wedge \psi$ and $s \nVdash \chi_2 \wedge \psi$, thus $\Theta_{2a}(s, t)$ which implies $\Theta_3(s, t)$. In the case where ϕ and ψ have multiple exact verifiers, we simply repeat the reasoning above. Therefore, any Γ that is not a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$ is also not a possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$, and since a set of formulas is either a possible $\mathcal{K}$ -subset or not, every possible $\mathcal{K}$ -subset under $\mathcal{EL}_3$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_4$.

Proof for (b): trivial.

Since the expansion with states does not always succeed, the expansion with formulas must also be given a new definition:

Definition 3.8 (Operations on Belief Set with Consistency Constraints). *Given a model M and a selection function γ , for any $\Sigma_n \subseteq \Sigma_1$,*

$$(\mathcal{EL}^\oplus) \ D_{\mathcal{EL}_n}(S^\mathcal{K}) \oplus_n \phi = \begin{cases} \gamma(D_{\mathcal{EL}_n}(S^\mathcal{K} \cup \{\sigma\}) \text{ s.t. } \sigma \Vdash \phi & \text{if } S^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ D_{\mathcal{EL}_n}(S^\mathcal{K}) & \text{otherwise} \end{cases}$$

$$(\mathcal{EL}^\ominus) \ D_{\mathcal{EL}_n}(S^\mathcal{K}) \ominus_{\mathcal{EL}_n} \phi = D_{\mathcal{EL}_n}(S^\mathcal{K} \setminus \{\sigma : \sigma \Vdash \phi\})$$

$$(\mathcal{IL}^\oplus) \ D_{\mathcal{IL}_n}(S^\mathcal{K}) \oplus_n \phi = \begin{cases} \gamma(D_{\mathcal{IL}_n}(S^\mathcal{K} \cup \{\sigma\}) \text{ s.t. } \sigma \Vdash \phi & \text{if } S^\mathcal{K} \cup \{\sigma\} \in \Sigma_n \\ D_{\mathcal{IL}_n}(S^\mathcal{K}) & \text{otherwise} \end{cases}$$

$$(\mathcal{IL}^\ominus) \ D_{\mathcal{IL}_n}(S^\mathcal{K}) \ominus_{\mathcal{IL}_n} \phi = D_{\mathcal{IL}_n}(S^\mathcal{K} \setminus \{\sigma : \sigma \Vdash \phi\})$$

$$(\mathcal{EL}^\otimes) \ D_{\mathcal{EL}_n}(S^\mathcal{K}) \otimes_{\mathcal{EL}_n} \phi = \gamma(D_{\mathcal{EL}_n}(S^\mathcal{K} * \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

$$(\mathcal{IL}^\otimes) \ D_{\mathcal{IL}_n}(S^\mathcal{K}) \otimes_{\mathcal{IL}_n} \phi = \gamma(D_{\mathcal{IL}_n}(S^\mathcal{K} * \sigma)) \text{ s.t. } \sigma \Vdash \phi$$

And for a reason similar to **Theorem 3.1**, expanding, contracting and revising with formulas on $\mathcal{K}$ under $\mathcal{EL}_3$ and $\mathcal{EL}_4$ are also equivalent. Also, it should be noticed that although the idea of Levi-identity is preserved in defining the revision with states, it does not hold in the revision with formulas. That is, not necessarily $(D_{\mathcal{EL}_n}(S^\mathcal{K}) \ominus_{\mathcal{EL}_n} \phi) \oplus_n \phi = D_{\mathcal{EL}_n}(S^\mathcal{K}) \otimes_{\mathcal{EL}_n} \phi$, and likewise for $\mathcal{IL}$ -series frameworks. However, the failure of Levi-identity is required, if one wishes revision to be always successful.

3.2.1 The “Spirit” of Paraconsistency

It is shown above that the selected frameworks with locality and consistency constraints satisfy **Local Rationality**: the modelled epistemic agent’s belief set would have non-trivial subject

matter; and they do not have a possible $\mathcal{K}$ -subset in the form of $\{\phi, \neg\phi\}$, and are closed under modus ponens. It should also be clear that the new frameworks do equally well in satisfying the desiderata which simple frameworks could satisfy, except **Paraconsistency**.

Whether **Paraconsistency** is satisfied under $\mathcal{EL}_3, \mathcal{EL}_4$, and $\mathcal{IL}_4$ may be disputed, not because they cannot admit an inconsistent belief set without explosion, but because they cannot admit a logically inconsistent belief set *at all*. While the consistency constraint for $\mathcal{EL}_1$ is too weak to forbid $p \vee \neg p \in \mathcal{K}$, the consistency constraints for the remainder are too strong, such that for any possible $\mathcal{K}$ -subset Γ , $\Gamma \not\vdash \perp$.

In theory, **Paraconsistency** is satisfied vacuously. Since the belief set can never be logically inconsistent, explosion also does not occur. Yet one might contend that those three theories do not capture the “spirit” of the desideratum, for the reason we employed **Paraconsistency** among our desiderata is that we wanted the modelled epistemic agent to deal with inconsistent belief without explosion, not that we wanted to forbid the agent to have inconsistent belief at all.

A primary answer is that we need to compromise **Local Rationality** and the alleged “spirit” of **Paraconsistency**. If one values the possibility of having logically inconsistent beliefs highly, one could prefer $\mathcal{EL}_1$, and permit the modelled epistemic agent to believe that $\phi \wedge \neg\phi$. If one holds that rationality must be strong enough to exclude contradictory beliefs like $\phi \wedge \neg\phi$, or a pair of beliefs that $\phi \wedge \psi$ and that $\neg\phi \wedge \neg\psi$, consistency constraints are just too coarse-grained to permit any other kinds of logical inconsistency. It, of course, permits an impossible belief set, i.e., there being no possible world in the model that makes every belief of the modelled agent true. Yet the complaint was about not permitting the belief set to be *logically* impossible, not metaphysically.

That being said, proponents of the $\mathcal{IL}$ -series might desire another kind of framework. If one holds that the belief set must be able to be logically inconsistent and that the belief set need not be closed under conjunction elimination, one can embrace framework $\mathcal{EL}_1$. But as mentioned above, there is only one kind of framework among the $\mathcal{IL}$ -series - $\mathcal{IL}_4$ —which is free from blatant inconsistency. If one holds that the belief set must be closed under conjunction elimination, the approach of employing consistency constraints on Σ cannot preserve the “spirit” of **Paraconsistency**, if one values it.

Given such considerations, one might desire approaches to **Local Rationality** other than consistency constraints.

3.3 Weighing and Cancellation

Yet there could be more motivations for alternatives to consistency constraints. For one thing, one may find the idea that an ideal epistemic agent must only have sources of belief which favor only one of ϕ or $\neg\phi$ (or nothing at all) not convincing. At least we humans do not behave like that. We sometimes have reasons for both ϕ and $\neg\phi$, and after some explicit or implicit consideration, we either favor one of them, or just suspend judgment and believe neither of them. One might favor an approach which decides the belief set from a belief-state set via similar procedures.

A related problem is that there seems to be no particularly good philosophical reason to think that expansion with states sometimes fails. Again, at least we humans have no problem entertaining a new source of belief that contradicts our current belief; and it is unclear why an

ideal agent should be disabled in this regard.

So an immediate idea is to modify the function Δ , so it can pick at most one formula between ϕ and $\neg\phi$ when $\mathcal{S}^\mathcal{K}$ comprises verifiers for both ϕ and $\neg\phi$. But which one? Perhaps it should be what one is more *entrenched* into.

We already defined entrenchment in **Definition 2.6**. In short, given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$, the entrenchment of a formula is the number of its verifiers in $\mathcal{S}^\mathcal{K}$. In a similar vein, we define the notions of *resistance* and *net entrenchment*:

Definition 3.9 (Entrenchment, Resistance, and Net Entrenchment). *Given a model M and a belief-state set $\mathcal{S}^\mathcal{K}$,*

($E_\mathcal{E}$) *The exact entrenchment of ϕ $E_\mathcal{E}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$, and*

($E_\mathcal{I}$) *The inexact entrenchment of ϕ $E_\mathcal{I}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \Vdash \phi\})$, and*

($R_\mathcal{E}$) *The exact resistance of ϕ $R_\mathcal{E}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \dashv\vdash \phi\})$, and*

($R_\mathcal{I}$) *The inexact resistance of ϕ $R_\mathcal{I}(\phi) = \text{card}(\mathcal{S}^\mathcal{K} \cap \{\sigma : \sigma \dashv\vdash \phi\})$.*

The net entrenchment is intuitively the resistance subtracted from entrenchment. But it should be noticed that both of them may be infinite. Thus:

$$\begin{aligned} \text{The exact net entrenchment of } \phi \text{ } NE_\mathcal{E}(\phi) &= \begin{cases} E_\mathcal{E}(\phi) - R_\mathcal{E}(\phi) & \text{if both are finite} \\ E_\mathcal{E}(\phi) & \text{if } E_\mathcal{E}(\phi) > R_\mathcal{E}(\phi) \text{ and } E_\mathcal{E}(\phi) \geq \aleph_0 \\ -(R_\mathcal{E}(\phi)) & \text{if } E_\mathcal{E}(\phi) < R_\mathcal{E}(\phi) \text{ and } R_\mathcal{E}(\phi) \geq \aleph_0 \\ b & \text{if } E_\mathcal{E}(\phi) = R_\mathcal{E}(\phi) \geq \aleph_0 \end{cases} \\ \text{The inexact net entrenchment of } \phi \text{ } NE_\mathcal{I}(\phi) &= \begin{cases} E_\mathcal{I}(\phi) - R_\mathcal{I}(\phi) & \text{if both are finite} \\ E_\mathcal{I}(\phi) & \text{if } E_\mathcal{I}(\phi) > R_\mathcal{I}(\phi) \text{ and } E_\mathcal{I}(\phi) \geq \aleph_0 \\ -(R_\mathcal{I}(\phi)) & \text{if } E_\mathcal{I}(\phi) < R_\mathcal{I}(\phi) \text{ and } R_\mathcal{I}(\phi) \geq \aleph_0 \\ b & \text{if } E_\mathcal{I}(\phi) = R_\mathcal{I}(\phi) \geq \aleph_0 \end{cases} \end{aligned}$$

where $\forall z \in \mathbb{Z}, b \not\geq z$ & $b \not\leq z$. Especially, $b \not\geq 1$.

Metaphorically, the net entrenchment of ϕ is the quantitative representation of the “weighing” of reasons for believing ϕ or $\neg\phi$. If there are more reason to believe that ϕ than to believe that $\neg\phi$, the net entrenchment would be positive; and negative if vice versa. If there is as much reason to believe that ϕ as to believe that $\neg\phi$, the net entrenchment would be 0 or “balanced”—and one has good reason to suspend judgment between ϕ and $\neg\phi$.

Then the belief set $\mathcal{K}$, given a belief-state set $\mathcal{S}^\mathcal{K}$, would be the set of all formulas that have positive net entrenchment. This gives us two more kinds of frameworks:

Definition 3.10 (Frameworks $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$). *Given a model M ,*

The framework $\mathcal{ELW}_0$ is a quintuple $\langle M, \Sigma_0, +_0, \div, \Delta_{\mathcal{ELW}} \rangle$ where $\Delta_{\mathcal{ELW}}(X) = \{\phi : NE_\mathcal{E}(\phi) \geq 1 \text{ \& } \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \square(\tau \sqsubseteq \text{sm}(\phi) \text{ \& } \tau \sqsubseteq \sqcup X)\} \subseteq \mathcal{L}$.³

The framework $\mathcal{ILW}_0$ is a quintuple $\langle M, \Sigma_0, +_0, \div, \Delta_{\mathcal{ILW}} \rangle$ where $\Delta_{\mathcal{ILW}}(X) = \{\phi : NE_\mathcal{I}(\phi) \geq 1 \text{ \& } \forall \psi \in \text{Sub}(\phi), \exists \tau \neq \square(\tau \sqsubseteq \text{sm}(\phi) \text{ \& } \tau \sqsubseteq \sqcup X)\} \subseteq \mathcal{L}$.

³If one wants to model a more “prudent” epistemic agent, one can adjust the clause $NE_\mathcal{E}(\phi) \geq 1$ by increasing the number of the right-handed side. Similarly for $\mathcal{ILW}_0$.

One interesting feature of $\mathcal{ELW}_0$ is that it has the same class of possible $\mathcal{K}$ -subsets as that of $\mathcal{EL}_1$.

Theorem 3.2 (Equivalence of Possible $\mathcal{K}$ -subset Under $\mathcal{ELW}_0$ and $\mathcal{EL}_1$). *(a) Every possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$ is a possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ and (b) every possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$.*

Proof for (a): We prove the stronger thesis that any belief set under $\mathcal{EL}_1$ can be a belief-set under $\mathcal{ELW}_0$. By definition, any belief set under $\mathcal{EL}_1$, $\Gamma_1 = \Delta_{\mathcal{EL}}(X)$ for some $X \in \Sigma_1$. Since $\Sigma_1 \subseteq \Sigma_0$, we get $X \in \Sigma_0$; and since, if there are some exact verifiers of ϕ and no exact falsifiers of ϕ in $S^\mathcal{K}$, $NE_\mathcal{E}(\phi) \geq 1$, we get $\Delta_{ELW}(X) = \Gamma_1$. Thus, any belief set under $\mathcal{EL}_1$ can be a belief-set under $\mathcal{ELW}_0$ in the exact same model; a fortiori, every possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$ is a possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$.

Proof for (b): Let Γ_0 be an arbitrary possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$. Given Γ_0 , we can construct a model M' which gives a unique exact verifier σ_ϕ for each $\phi \in \Gamma_0$ unless ϕ is a disjunction. Then we check whether $\Gamma_0 \subseteq \mathcal{K} = \Delta_{\mathcal{EL}}(S^\mathcal{K})$ for some $S^\mathcal{K} \in \Sigma_1$. By definition of Δ_{EL} , $\Gamma_0 \subseteq \Delta_{EL}(S^\mathcal{K})$ if and only if $S^\mathcal{K}$ comprises σ_ϕ for all $\phi \in \Gamma_0$. Let $S^{\mathcal{K}'}$ denote the smallest possible $S^\mathcal{K}$ that comprises σ_ϕ for all $\phi \in \Gamma_0$, i.e., $S^{\mathcal{K}'} = \{\sigma_\phi : \phi \in \Gamma_0\}$. Thus we check whether such $S^{\mathcal{K}'}$ can satisfy the following two conditions: (1) $S^{\mathcal{K}'} \subseteq S$, (2) $\forall s, t \in S^{\mathcal{K}'}, \neg \exists \psi (s \Vdash \psi \ \& \ t \dashv \Vdash \psi)$. Since by supposition, $\forall \phi \in \Gamma_0, \sigma_\phi \in S$, (1) can be satisfied. For (2), suppose, for contradiction, that there is some χ whose exact verifier and falsifier are in $S^\mathcal{K}$, which makes $\{\chi, \neg\chi\} \subseteq \Delta_{\mathcal{EL}}(S^{\mathcal{K}'})$. Then, by supposition, $\{\chi, \neg\chi\} \subseteq \Gamma_0$ which yields that $NE_\mathcal{E}(\chi) \geq 1$ and $NE_\mathcal{E}(\neg\chi) \geq 1$, and therefore $E_\mathcal{E}(\chi) > R_\mathcal{E}(\chi)$ and $E_\mathcal{E}(\neg\chi) > R_\mathcal{E}(\neg\chi)$. But since by definition $E_\mathcal{E}(\chi) = R_\mathcal{E}(\neg\chi)$ and $R_\mathcal{E}(\chi) = E_\mathcal{E}(\neg\chi)$, we get $E_\mathcal{E}(\chi) > R_\mathcal{E}(\chi)$ and $R_\mathcal{E}(\chi) > E_\mathcal{E}(\chi)$, contradiction. Thus (2) is also satisfied, and therefore $S^{\mathcal{K}'} \in \Sigma_1$, given the canonical model M' . Therefore, Γ_0 is also a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$, and since there was no specific supposition on Γ_0 , every possible $\mathcal{K}$ -subset under $\mathcal{ELW}_0$ is a possible $\mathcal{K}$ -subset under $\mathcal{EL}_1$.

Operations with states in $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ are in many aspects similar to simple frameworks. Expansion always succeeds, and revision is *not* defined. Since there are no consistency constraints imposed on Σ_0 , it will be artificial to decide which states must be contracted before expanding with a state in order to revise with it.

Yet it is not an obvious defect, as it was in simple frameworks. In some cases, expansion alone is enough to model revision, in the case where entrenchment and resistance are finite. Suppose $S^\mathcal{K}$ is such that $E_\mathcal{I}(\phi) = 1$ and $R_\mathcal{I}(\phi) = 2$, and σ_1 and σ_2 are both inexact verifiers of ϕ not in $S^\mathcal{K}$. Then we get $\neg\phi \in \Delta_{\mathcal{ILW}}(S^\mathcal{K})$, $\neg\phi \notin \Delta_{\mathcal{ILW}}(S^\mathcal{K} + \sigma_1)$ and $\phi \in \Delta_{\mathcal{ILW}}((S^\mathcal{K} + \sigma_1) + \sigma_2)$. So although it may sound strange, the *expansion* postulate on expansion (see Table 1) does not hold under $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$.

The operations with formulas are also identical to those of simple frameworks (see **Definition 2.5**). One intriguing feature of the two new frameworks is that contracting with a formula that is not in $\mathcal{K}$ can still have substantial impact on $\mathcal{K}$. Also, here expanding with formulas does not always succeed if the net entrenchment is small enough, contrary to the expansion with states. One might wish to modify the definition to guarantee success, for example, by defining expansion with ϕ as the decided belief set after expanding with a sufficient number of verifiers of ϕ in $S^\mathcal{K}$. However, this line of definition is either insufficient to guarantee success, or leads to severe ill-definition. For one thing, the model given might just supply us with an

insufficient number of verifiers compared to falsifiers. Suppose a belief-state set is given such that $E_{\mathcal{E}}(\phi) = 1$ and $R_{\mathcal{E}}(\phi) = 5$, yet the model gives only one exact verifier of ϕ in S . Then there is just not enough number of verifiers of ϕ to force the expansion to succeed. Therefore, as far as I can think of, the present definition is as good as it gets.⁴

Besides operations and inconsistency, another notable feature of frameworks with weighing is that it permits the modelled agent to believe $\phi \vee \psi$ without believing either of them. In simple frameworks and frameworks with consistency constraints, the modelled agent must believe either of the disjuncts to believe a disjunction, since a verifier of a disjunction is by definition a verifier of either of its disjuncts. But this may be contended, for it seems clearly possible for an ideal agent to believe a disjunction without being committed to either of its disjuncts. It is particularly so when we consider $\phi \vee \neg\phi$. I do not have the belief that it will rain a week later, nor that it will not rain a week later, but I have the belief that either it will rain a week later or not.

$\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ neatly account for this kind of case. Suppose $E_{\mathcal{E}}(\phi) = R_{\mathcal{E}}(\phi) = E_{\mathcal{E}}(\psi) = R_{\mathcal{E}}(\psi) = 1$, and $E_{\mathcal{E}}(\neg\phi \wedge \neg\psi) = 3$. Then, although both ϕ and ψ would not be included in $\mathcal{K}$ for their net entrenchment is zero, $NE_{\mathcal{E}}(\phi \vee \psi) = 1$ and therefore $\phi \vee \psi \in \mathcal{K}$. Likewise for $\mathcal{ILW}_0$. And this is also intuitively telling: even if one has equally many sources to believe ϕ and $\neg\phi$, ψ and $\neg\psi$ and therefore rejects believing either of them, if one has less reason to think neither ϕ nor ψ is the case than the sum of the reasons to believe that ϕ and that of ψ , one should reasonably believe that either ϕ or ψ would be the case.

However, one unfortunate consequence of both frameworks is that it does not yield the closure of $\mathcal{K}$ under modus ponens. It is not because it permits $\{\phi, \neg\phi \vee \psi, \neg\phi\} \subseteq \mathcal{K}$, but because it permits beliefs of disjunctions without disjuncts, as mentioned above. Suppose $NE_{\mathcal{E}}(\phi) = 1$, $NE_{\mathcal{E}}(\neg\psi) = 0$, and $E_{\mathcal{E}}(\neg\phi \wedge \psi) = 0$. Then although $NE_{\mathcal{E}}(\neg\phi \vee \psi) \geq 1$, $NE_{\mathcal{E}}(\psi) = 0$ and thus $\{\phi, \phi \vee \neg\psi\} \subseteq \mathcal{K}$ but $\psi \notin \mathcal{K}$. Likewise for $\mathcal{ILW}_0$.

Of course, one may argue that therefore the closure under modus ponens was not desirable at all. I am personally sympathetic to this view, but one may simply favor frameworks with consistency constraints.

Before moving to the conclusion, let me lay out a rough idea for a final kind of framework. One might think that if the modelled epistemic agent decided to believe ϕ after “weighin”, the agent must “cancel out” the states in $\mathcal{S}^{\mathcal{K}}$ that falsify ϕ , so that the cancelled states should not further affect reasoning for more complex formulas.

This is motivated by the problem that under every framework developed so far, conjunctions are too difficult to be believed. I mentioned in §1.1.2 that there may be cases where one believes ϕ and ψ but not $\phi \wedge \psi$. Nevertheless, in most cases, equipped with the belief that ϕ and ψ , the belief that $\phi \wedge \psi$ should not be too difficult to acquire. However, even if the belief-state set contains exact verifiers for both ϕ and ψ , that helps nothing at all in believing that $\phi \wedge \psi$. The case is even worse in frameworks with weighing, since even if there are more exact verifiers for the conjuncts than their exact falsifiers, while the exact verifiers do not help with the entrenchment of the conjunction, the exact falsifiers do help with the resistance of the conjunction.

⁴The present definition also risks being ill-defined, to be fair, if one permits the set of verifiers of a formula to be empty. But recall that the primary definition of a model employed in our paper does not permit the case (see **Definition 1.6**). If one permits an empty set of verifiers, one can add a clause to the definition stating that if ϕ has no verifier, expanding with ϕ on $\mathcal{K}$ does not cause any difference.

The initial idea is as follows. Given the language $\mathcal{L}$, we first sort the formulas by their depth to construct $\mathcal{L}' = \langle \Gamma^0, \Gamma^1, \Gamma^2, \dots \rangle$ where $\Gamma^n = \{\phi : \phi \in \mathcal{L} \text{ \& } \text{depth}(\phi) = n\}$. Now, given a model M and an initial belief-state set $\mathcal{S}^\mathcal{K}$ (for convenience, let $\mathcal{S}^\mathcal{K} = \mathcal{S}^{\mathcal{K}_0}$), let $\mathcal{K}_0 = \Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}_0})$ (or $\Delta_{\mathcal{ELW}}(\mathcal{S}^\mathcal{K})$). I will lay out the idea with the inexact notion of verification here). From $\mathcal{K}_0$ we construct a new belief-state set after cancelling all the “defeated” states in $\mathcal{S}^\mathcal{K}$, that is, $\mathcal{S}^{\mathcal{K}_1} = \mathcal{S}^{\mathcal{K}_0} \setminus \{\sigma : \exists \phi \in \Gamma^0 (\phi \in \mathcal{K}_0 \wedge \sigma \triangleleft \phi)\}$. With $\mathcal{S}^{\mathcal{K}_1}$ we re-decide a belief set $\mathcal{K}_1 = \Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}_1})$, and construct $\mathcal{S}^{\mathcal{K}_2} = \mathcal{S}^{\mathcal{K}_1} \setminus \{\sigma : \exists \phi \in \Gamma^1 (\phi \in \mathcal{K}_1 \wedge \sigma \triangleleft \phi)\}$, and so on. In general, for any $n > 0$, $\mathcal{K}_n = \Delta_{\mathcal{ELW}}(\mathcal{S}^{\mathcal{K}_n})$ and $\mathcal{S}^{\mathcal{K}_n} = \mathcal{S}^{\mathcal{K}_{n-1}} \setminus \{\sigma : \exists \phi \in \Gamma^n (\phi \in \mathcal{K}_{n-1} \wedge \sigma \triangleleft \phi)\}$. Under the new frameworks with cancellation, then, the belief set would be identified as $\mathcal{K}^\infty$.

But this crude idea is not satisfying. For one thing, we might wish $\mathcal{K}_0$ and $\mathcal{K}_1$ to agree with regard to atomic formulas, $\mathcal{K}_1$ and $\mathcal{K}_2$ to agree with regard to formulas whose depth is 1, and so on. Yet with such a primeval mechanism one cannot guarantee the equivalence. Suppose $\Gamma^0 = \{p, q\}$, and the model gives that $|p| = \langle \{s_1, s_2\}, \{t_1\} \rangle$ and $|q| = \langle \{s_1, s_2\}, \{t_1, t_2\} \rangle$, and $\{s_1, s_2, t_1, t_2\} \subseteq \mathcal{S}^\mathcal{K}$. Since $NE_\mathcal{E}(p) = 1$, $p \in \mathcal{K}_0$ and therefore $t \notin \mathcal{S}^{\mathcal{K}_0}$. Then although $NE_\mathcal{E}(q) = 0$ and therefore $q \notin \mathcal{K}_0$, since $\mathcal{S}^{\mathcal{K}_1}$ comprises two exact verifiers of q and one exact falsifier of q , $q \in \mathcal{K}_1$. There are two ways this may be problematic: either because one would not wish new formulas with depth n to appear at $\mathcal{K}_{n+1}$ which were not in $\mathcal{K}_n$; or because one might wish that, if new formulas with depth n occur after cancelling out from $\mathcal{S}^{\mathcal{K}_n}$, the falsifiers for the new formulas must be cancelled out from future weighing.

Another trickier problem to solve is that if we define the cancellation process as above, the pleasing feature of $\mathcal{ELW}_0$ and $\mathcal{ILW}_0$ that the belief set may contain a disjunction without containing any of its disjuncts would be weaker. It would be easy to check that under those two frameworks, it was possible to believe that $\phi \vee \psi$ alongside the belief that $\neg\phi$ and $\neg\psi$. This, however, would be impossible with the crude idea, since the falsifier of $\neg\phi$ and $\neg\psi$ would be the verifier of $\phi \vee \psi$, and therefore if some $\mathcal{K}^n$ consists of $\neg\phi$ and $\neg\psi$, it would clear all the verifiers of $\phi \vee \psi$ at the next cancellation process. This worry, of course, may be relieved by a philosophical argument that it is not desirable to allow the ideal epistemic agent to believe that $\phi \vee \psi$ alongside the belief that $\neg\phi$ and $\neg\psi$. However, at least I do not have one.

Still, I believe that the idea of cancellation may be worth spending time developing, although it may be quite difficult to flesh out a satisfying construction.

4 Conclusion

In this paper, I developed a family of hyperintensional frameworks for belief revision by utilizing the resources of Finean truthmaker semantics. All of them (nominally, at least) satisfy **Definite Contraction**, **Logical Ignorance**, **Paraconsistency**, **Hyperintensionality**, and **Pertinent Recovery**; while **Local Rationality** is satisfied by $\mathcal{EL}_1$, $\mathcal{EL}_3$, $\mathcal{EL}_4$, $\mathcal{IL}_4$, $\mathcal{ELW}_0$, $\mathcal{ILW}_0$ with different degrees and means.

I intentionally did not develop any arguments favoring particular frameworks among them; partially because it exceeds the scope of this paper, but also because, along the truthmaker-semantic approach, one might be able to develop better frameworks than what is proposed in this paper.

Let me finally note one general limitation of the approach. Even in frameworks $\mathcal{ELW}_0$, $\mathcal{ILW}_0$, for the modelled epistemic agent to believe that $p \vee \neg p$, it must have non-zero entrenchment in either p or $\neg p$. But this idea may be challenged, since one might believe $p \vee \neg p$

because one knows that this is just logically true, without having any sources to believe either p or $\neg p$. And with **Definition 1.7**, it is not clear at all how to permit the mentioned case without infringing **Logical Ignorance**.

I believe that the problem should be dealt with by expanding our language to a higher-order language. Then one can model the case where one believes that $\forall\phi(\phi \vee \neg\phi)$, every verifier of which inexactly verifies $p \vee \neg p$, but generally not p nor $\neg p$. The remaining task, then, is to develop a truthmaker semantics that can accommodate higher-order language, which can be employed to construct a framework of belief-revision with higher-order language.

References

- Alchourrón, C. E., Gärdenfors, P., & Makinson, D. (1985). On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2), 510–530. doi: 10.2307/2274239
- Correia, F., & Skiles, A. (2017). Grounding, essence, and identity. *Philosophy and Phenomenological Research*, 98(3), 642–670. doi: 10.1111/phpr.12468
- Fine, K. (2012). The pure logic of ground. *Review of Symbolic Logic*, 5(1), 1–25. doi: 10.1017/s1755020311000086
- Fine, K. (2016). Angellic content. *Journal of Philosophical Logic*, 45(2), 199–226. doi: 10.1007/s10992-015-9371-9
- Fine, K. (2017a). A theory of truthmaker content i: Conjunction, disjunction and negation. *Journal of Philosophical Logic*, 46(6), 625–674. doi: 10.1007/s10992-016-9413-y
- Fine, K. (2017b). A theory of truthmaker content ii: Subject-matter, common content, remainder and ground. *Journal of Philosophical Logic*, 46(6), 675–702. doi: 10.1007/s10992-016-9419-5
- Fine, K. (2017c). Truthmaker semantics. In B. Hale, C. Wright, & A. Miller (Eds.), *A companion to the philosophy of language* (pp. 556–577). Wiley-Blackwell.
- Gärdenfors, P., & Makinson, D. (1988). Revisions of knowledge systems using epistemic entrenchment. In *Proceedings of the 2nd conference on theoretical aspects of reasoning about knowledge* (pp. 83–95).
- Hansson, S. O. (2022). Logic of Belief Revision. In E. N. Zalta (Ed.), *The Stanford encyclopedia of philosophy* (Spring 2022 ed.). Metaphysics Research Lab, Stanford University. <https://plato.stanford.edu/archives/spr2022/entries/logic-belief-revision/>.
- Jago, M. (2024). Belief in truthmaker semantics. *Synthese*, 204(4), 1–24. doi: 10.1007/s11229-024-04748-z
- van Ditmarsch, H., van der Hoek, W., & Kooi, B. (2008). *Dynamic epistemic logic*. Dordrecht, Netherland: Springer.